

12.421/12.621 Lecture 1: Material Properties

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Topics covered

- Constitutive parameters
- Permittivity and material polarization
- Simple models for the permittivity of water and water ice

1 Inferences from remote sensing data

- It is often the case that we want to infer physical properties of observed areas on Earth or other bodies from remote sensing (RS) data.
- In this course (and in general), we have defined RS methods as those that utilize electromagnetic (EM) radiation.
- The EM radiation of interest in this course is either emitted from, or has interacted with, the medium of interest.
- Our goal throughout most of this course, then, is to develop a foundational understanding of and intuition for the interaction of EM radiation with matter.
- This will be done through several broad topics, starting with, in order:
 1. Constitutive parameters
 2. Classical EM principles (Maxwell's equations)
 3. Wave propagation
- Today, we cover constitutive parameters.

2 Constitutive parameters

The EM characteristics of any material can be described by four constitutive parameters:

1. ★ $\epsilon' \epsilon_0$ is the electrical *permittivity*
 - measure of a material's ability to store an electric charge
 - units: F/m, where F = Farad = J/V² is a unit of capacitance
 - written above as the product of two parameters:
 - $\epsilon_0 = 8.85 \times 10^{-12}$ F/m is the permittivity of free space
 - $\epsilon' = \epsilon_{\text{material}}/\epsilon_0$ is the (dimensionless) relative permittivity (relative to free space)
 - it is almost always the case that $\epsilon' \geq 1$ (exceptions discussed later)

- ϵ' is of *fundamental importance in RS* and is something we delve into more in this lecture

2. q_v is the *charge density*

- units: C/m³, where C = Coulomb = A·s
- $q_v = 0$ for RS applications, so it is not considered further in this course

3. μ is the magnetic *permeability*

- measure of a material's ability to support the development of a magnetic field
- units: H/m, where H = Henry = J/A² is a unit of inductance
- $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space
- $\mu = \mu_0$ in virtually all RS applications, so it will not be considered further in this course
- $c = 1/\sqrt{\mu_0\epsilon_0}$ is the speed of light in a vacuum

4. ★ σ is the *conductivity*

- measure of a material's ability to allow current to flow through it
- accounts for EM wave attenuation in materials, and is therefore of *fundamental importance in RS*
- units: S/m, where S = Siemens = Ω^{-1} = A/V
- general categories and examples:
 - $\sigma = 0$: “lossless” (e.g., vacuum)
 - $\sigma \approx 0$: “low-loss” (e.g., air)
 - $\sigma > 0$: “lossy” (e.g., seawater)
 - $\sigma \gg 1$: “conductor” (e.g., copper, gold)
 - $\sigma \rightarrow \infty$: “perfect conductor”

2.1 The complex dielectric constant

For RS applications, $\mu \approx \mu_0$ and $q_v = 0$, so we are mainly concerned with $\epsilon'\epsilon_0$ and σ . We combine these into a single, complex-valued parameter, the (dimensionless, complex) *dielectric constant*

$$\epsilon = \epsilon' - j\epsilon'', \quad \text{where} \quad \epsilon'' = \frac{\sigma}{\omega\epsilon_0} \quad (j^2 = -1) \quad (1)$$

and ω is the (angular) frequency of the EM wave. The derivation of ϵ will be done later in the course. Here $\sigma/(\omega\epsilon_0)$ is the contribution of conduction to ϵ'' . Polarization (relaxation) mechanisms also add to ϵ'' , as we will see in Section 5.

Comments on ϵ :

- ϵ' has already been discussed (relative permittivity).
- ϵ'' is the *loss factor*, which represents attenuation of the EM wave.
 - In general, RS applications involve narrow ranges of ω (and ϵ_0 is a universal constant).
 - Therefore, conduction losses are governed by the conductivity σ .
 - The absorbed energy is converted to heat; this is how microwave ovens work (through both the dipolar-relaxation and the ionic-conduction losses of water).

- Taking a broader view of RS, where the range of ω is large, we have $\varepsilon'(\omega)$ and $\sigma(\omega)$.
- Several mechanisms by which ε' and σ vary with ω will be covered throughout the course, but these can be classified mainly as stretching and rotating of atoms and molecules.
 - Going through these will give some physical insight into the surprisingly complicated transparency of the atmosphere at different EM frequencies.
- For now, let's keep things simple and build some intuition.

3 Electric fields in matter

3.1 Conduction

Conduction is something that everyone should have some broad understanding of: it is simply the ability of a material to allow the flow of electric current. Electric current is simply the movement of charged particles: electrons and/or ions.

- So, conduction is a description of the ability of electrons and/or ions to move in a material.
- Metals, of course, are examples of excellent conductors.
- In RS applications, seawater is the ubiquitous example of a highly conductive natural material.
 - We might therefore expect the loss factor ε'' to be high in seawater (more on this in a bit).

3.2 Permittivity

In dielectric materials (i.e., insulators), charges are attached to specific atoms or molecules and are not free to “roam.”

- Charges can only move *within* the atom or molecule.
- The cumulative effect of these small motions accounts for the characteristic behavior of dielectrics: permittivity.
- Remember that these motions are simply rotation and extension.

3.2.1 Parallel-plate capacitor

Let's dig into this a bit more by considering the simplest case of a parallel-plate capacitor.

- Suppose we have two conductive plates of area A separated by a distance h and a perfect vacuum.
- Let's give the top plate a positive charge $+Q$ and the bottom plate a negative charge $-Q$ (Figure 1).
- The positively charged plate produces an electric field $E_+ = Q/(2A\varepsilon_0)$ that points away from the upper plate.
- The negatively charged plate does the opposite.
- Thus, there is a net field $E = Q/(A\varepsilon_0)$ between the two plates ($E = 0$ outside the plates).
- The potential between the plates is then $V = Qh/(A\varepsilon_0)$.

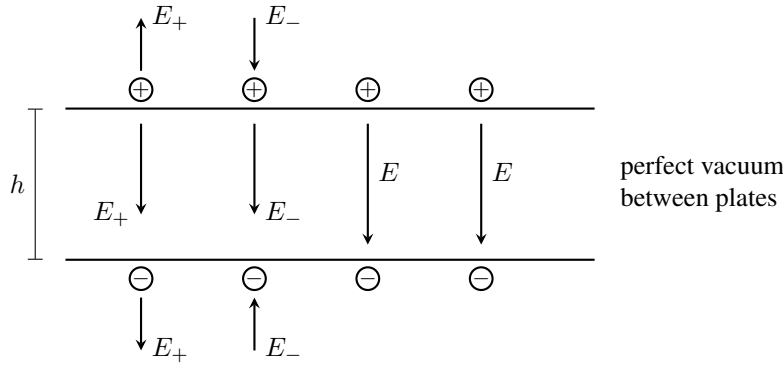


Figure 1: Side view of a parallel-plate capacitor with a perfect vacuum between the plates. The fields from the two plates add between the plates and cancel outside.

- The capacitance is simply

$$C = \frac{Q}{V} = \frac{A\epsilon_0}{h}. \quad (2)$$

We can see from Eq. (2) that $C \propto \epsilon_0$. Recall that we placed a perfect vacuum between our plates, so we might suppose that the general relation is $C \propto \epsilon'\epsilon_0$.

- Recalling that $\epsilon' \geq 1$, we see that increasing permittivity increases capacitance.
 - Countless observations back this up.
 - You may have personal experience with this, as commercial capacitors always come with some insulator between the conductors.

3.2.2 Parallel-plate capacitor with a dielectric

To see why this is, let's fill our parallel-plate capacitor with a dielectric.

- Let's choose a simple, ubiquitous dipole molecule: H_2O .
- Because H_2O is asymmetric, with its two H atoms oriented at an angle of 104.5° from one another, it behaves as an electric dipole with its own field $E_{\text{H}_2\text{O}}$ (Figure 2a).
- In the absence of an external electric field, H_2O molecules are randomly oriented.
- If we place these between the plates of our capacitor, they orient themselves with the electric field of the capacitor (Figure 2b).
- $E_{\text{H}_2\text{O}}$ opposes E , which reduces the *net* field.
 - Electrical energy is stored in the “polarization” (orientation of molecules) of the dielectric.

This is *permittivity* in a nutshell.

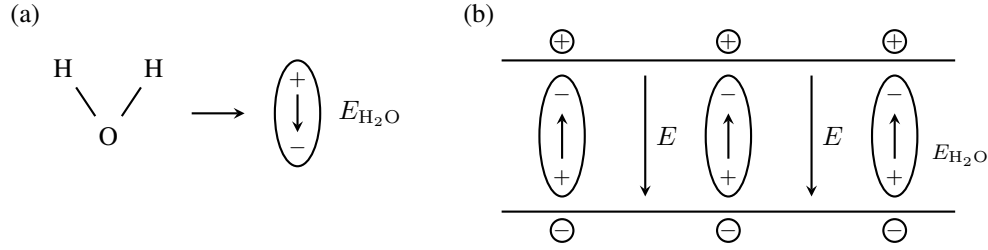


Figure 2: (a) The H_2O molecule acts as an electric dipole with field $E_{\text{H}_2\text{O}}$. (b) Between the plates of a capacitor, the dipoles align with the capacitor's field E , and $E_{\text{H}_2\text{O}}$ opposes E .

3.2.3 What the simple example leaves out

This is a highly simplified example meant to show the fundamental concepts. There are several important elements missing:

1. **Heat.** Vibrations associated with heat make perfect alignment of atoms or molecules impossible. Permittivity will therefore, in general, depend on temperature.
2. **Impurities.** Rarely in RS will you look at pure materials. Impurities or other inclusions have major influences on the effective permittivity. For example:
 - electrolytic solutions like salt water, where ionic salts increase conductivity by providing charges (ions) that can move large distances, producing an electric current;
 - heterogeneous mixtures like wet or dry soil and snow.
3. **Dynamics.** The previous example was static, but in RS we care about the response of materials to *oscillating* electric fields.
 - Because the field is oscillating, we need to consider the response time of atoms and molecules to the electric field. We will call this response time the “relaxation time.”
 - Different mechanisms that influence permittivity have different relaxation times, so the dominant mechanism is governed by ω :
 - *electronic or atomic polarization*: displacement of electron orbits relative to the nucleus $\rightarrow$ highest frequencies
 - *ionic polarization*: displacement of ion bonds in crystals $\rightarrow$ high frequencies
 - *dipolar polarization* (just discussed in the capacitor example): orientation of molecular dipoles $\rightarrow$ low frequencies
 - *interfacial (Maxwell–Wagner) polarization*: build-up of charge at boundaries between grains or materials with different electrical properties $\rightarrow$ lowest frequencies

3.3 A simple model for the response of materials to oscillating fields

Here we approximate the response of materials to oscillating fields using classical mechanics.

- To build some intuition, let us consider that electrons in an atom, molecules in a crystal, and dipoles in a mixture are bound by a force that can be treated (for small displacements) as a spring with constant K_{spring} :

$$F_{\text{bond}} = -K_{\text{spring}} \xi = -m\omega_0^2 \xi, \quad (3)$$

where m is the mass of our particle and ω_0 is the natural frequency.

- Let us also suppose there will be some damping mechanism (e.g., a friction-like force as in our capacitor example above, radiation in the case of atomic excitations, etc.; all will be covered in due course during the term). Damping must act opposite the velocity of the particle, and we *assume* (to be justified later in the course) that it is proportional to velocity:

$$F_{\text{damping}} = -m\nu \frac{d\xi}{dt}. \quad (4)$$

- The driving force is related to our oscillating electric field:

$$F_{\text{driving}} = qE(t) = q \operatorname{Re} [\tilde{E} e^{j\omega t}], \quad (5)$$

where q is the charge and $\tilde{E}$ is the complex amplitude (phasor) of the oscillating field.

- Newton's second law gives

$$m \frac{d^2 \xi}{dt^2} = F_{\text{bond}} + F_{\text{damping}} + F_{\text{driving}} \Rightarrow \frac{q}{m} E(t) = \frac{d^2 \xi}{dt^2} + \nu \frac{d\xi}{dt} + \omega_0^2 \xi, \quad (6)$$

which is a damped harmonic oscillator.

- This equation has a solution $\xi(t) = \operatorname{Re}[\tilde{\xi} e^{j\omega t}]$, where $\tilde{\xi}$ is the complex amplitude (*phasor*) of the displacement (phasors are explained in a box in Lecture 2, Section 3.2). Because the equation is linear, each time derivative becomes a factor $j\omega$ acting on the phasors, so

$$\frac{q}{m} \tilde{E} = -\omega^2 \tilde{\xi} + j\omega\nu \tilde{\xi} + \omega_0^2 \tilde{\xi}. \quad (7)$$

Solving for $\tilde{\xi}$ yields

$$\tilde{\xi} = \frac{(q/m) \tilde{E}}{\omega_0^2 - \omega^2 + j\omega\nu}. \quad (8)$$

- Let us define the (complex amplitude of the) dipole moment $\tilde{p} = q\tilde{\xi}$ and the polarization per unit volume

$$\tilde{P} = N\tilde{p} = Nq\tilde{\xi}, \quad (9)$$

where N is the number of dipoles per unit volume.

- In all cases encountered in RS, we have $\tilde{P} = \epsilon_0(\epsilon - 1)\tilde{E}$. From Eq. (8), we can then solve for the dielectric constant as

$$\boxed{\epsilon(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 + j\omega\nu}} \quad (10)$$

where $\omega_p^2 = Nq^2/(\epsilon_0 m)$ defines a constant frequency ω_p .

- Recalling our earlier definition of the dielectric constant, $\epsilon(\omega) = \epsilon'(\omega) - j\epsilon''(\omega)$, we can write the permittivity ϵ' and loss factor ϵ'' as

$$\epsilon' = 1 + \frac{\omega_p^2(\omega_0^2 - \omega^2)}{(\omega^2 - \omega_0^2)^2 + (\omega\nu)^2}, \quad (11)$$

$$\epsilon'' = \frac{\omega_p^2 \omega \nu}{(\omega^2 - \omega_0^2)^2 + (\omega\nu)^2}. \quad (12)$$

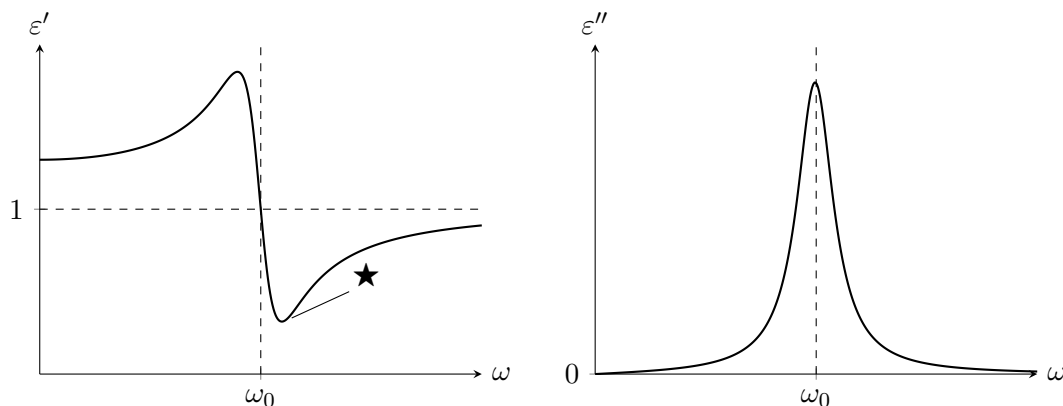


Figure 3: Permittivity ε' (left) and loss factor ε'' (right) of the damped harmonic oscillator model, Eq. (10).

Two points jump out of these equations (Figure 3):

1. $\varepsilon' = 1$ when $\omega = \omega_0$.
2. The maximum value of ε'' occurs (approximately, for small ν) at $\omega = \omega_0$.

★ Note that $\varepsilon' < 1$ just above the resonance frequency, even though we mentioned before that $\varepsilon' \geq 1$. Is this a problem? Not really. First, our model is highly simplified, and other mechanisms can act to increase ε' in real materials. Second, we'll cover this later, but suffice to say for now that energy travels at a maximum speed, and the above does not interfere with that.

4 Properties of natural materials

- The preceding discussion is a simplified model meant to build intuition. Like any classical approach to quantum phenomena, it is an approximation, but it turns out to be a really good one. More rigorous approaches using quantum mechanics yield similar results for the form of $\varepsilon(\omega)$.
- It's also worth pointing out that you can take essentially the same approach and then apply Ohm's law to get an expression for the conductivity $\sigma(\omega)$.
- Dependence on physical properties like temperature T is implicit in the previous model. This will come into the damping coefficient ν and the natural frequency ω_0 .

4.1 Temperature dependence

- *Permittivity.* We discussed before how atomic vibrations (heat) influence the polarizability (alignment of charges along the electric field) of materials. Permittivity is a description of polarizability, and thus will be a function of T .
- *Conductivity.* The T -dependence of conductivity varies depending on the material, primarily through how the number of free charge carriers (electrons or ions) and the rate at which they scatter change with temperature. In metals, the number of free electrons is essentially fixed, and stronger lattice vibrations (phonons) at higher T scatter them more often, so σ decreases with increasing T . In semiconductors and electrolytic solutions like seawater, higher T frees more carriers or makes ions more mobile, so σ increases with T . We won't get into details in this course, but suffice to say that

conductivity will have some T -dependence. The classic example is for metals, where electrical and thermal conductivity are linked by the Wiedemann–Franz law (cf. Drude theory)

$$\sigma = \frac{K}{LT}, \quad (13)$$

where

- K is the thermal conductivity,
- $L = \frac{3}{2} \left(\frac{k_B}{e} \right)^2 \approx 1.11 \times 10^{-8} \text{ W } \Omega/\text{K}^2$ is the Lorenz number from classical Drude theory (note: the quantum-mechanical (Sommerfeld) treatment gives $L = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 \approx 2.44 \times 10^{-8} \text{ W } \Omega/\text{K}^2$, which agrees better with measurements for most metals),
- k_B is the Boltzmann constant, and
- e is the electron charge.

This relation *only* holds for metals but, again, we should expect that conductivity depends on T for all materials.

5 Practical example: H₂O as pure liquid, liquid with salt, and solid

In RS applications, it is common to focus on specific frequency ranges corresponding to the instrument, and to consider dielectric properties in terms of a characteristic relaxation time τ_{rel} corresponding to a specific polarization mechanism. The relaxation time depends on the damping ν of the oscillator model above. For free charges (no restoring force, $\omega_0 = 0$), $\tau_{\text{rel}} = 1/\nu$, the mean time between collisions. For bound charges in the heavily damped limit ($\nu \gg \omega_0, \omega$), the oscillator model takes the Debye form below with $\tau_{\text{rel}} \approx \nu/\omega_0^2$.

5.1 Pure (distilled) water at microwave frequencies ($\omega/2\pi \leq 50 \text{ GHz}$)

- The most common model used in RS (at microwave frequencies) is the *Debye model* (named for Peter Debye). This takes a form very similar to the model we derived for harmonic oscillators:

$$\varepsilon = \varepsilon_{w\infty} + \frac{\varepsilon_{w0} - \varepsilon_{w\infty}}{1 + j\omega\tau_w} = \varepsilon' - j\varepsilon'', \quad (14)$$

where

- $\varepsilon_{w\infty} = \varepsilon(\omega \rightarrow \infty)$ is the high-frequency limit,
- $\varepsilon_{w0} = \varepsilon(\omega = 0)$ is the nominal or low-frequency limit (note that $\varepsilon_{w0} \neq \varepsilon_0$), and
- τ_w is the relaxation time.
- We can break this into real and imaginary components (note the similarity with the harmonic model):
$$\varepsilon' = \varepsilon_{w\infty} + \frac{\varepsilon_{w0} - \varepsilon_{w\infty}}{1 + (\omega\tau_w)^2}, \quad \varepsilon'' = \frac{\omega\tau_w(\varepsilon_{w0} - \varepsilon_{w\infty})}{1 + (\omega\tau_w)^2}. \quad (15)$$
- T -dependence comes into the model through ε_{w0} and τ_w , which are invariant over the given range of frequencies ($\omega/2\pi \leq 50 \text{ GHz}$).
 - $\varepsilon_{w0}(T)$ and $\tau_w(T)$ are empirical, commonly given as third-order polynomials in T . More on this in problem set 1.
 - $\varepsilon_{w\infty}$ is weakly dependent on T ; experiments show $\varepsilon_{w\infty} \approx 4.9$ for all T .
- See the plot and further notes on slide 4 of the lecture slides.

5.2 Salt water

- To account for the effects of ionizing salt on the dielectric constant of water, it is common to use a so-called *double-Debye* dielectric model.
- This model takes into account multiple polarizing mechanisms and the conductivity allowed by the presence of salt:

$$\varepsilon' = \varepsilon_{w\infty} + \frac{\varepsilon_{w0} - \varepsilon_{w1}}{1 + (\omega\tau_{w1})^2} + \frac{\varepsilon_{w1} - \varepsilon_{w\infty}}{1 + (\omega\tau_{w2})^2}, \quad (16)$$

$$\varepsilon'' = \frac{\omega\tau_{w1}(\varepsilon_{w0} - \varepsilon_{w1})}{1 + (\omega\tau_{w1})^2} + \frac{\omega\tau_{w2}(\varepsilon_{w1} - \varepsilon_{w\infty})}{1 + (\omega\tau_{w2})^2} + \frac{\sigma}{\omega\varepsilon_0}, \quad (17)$$

where ε_0 is the permittivity of free space (reminder: $\varepsilon_{w0} \neq \varepsilon_0$).

- In this model, ε_{w1} is ε at some intermediate frequency (details not important here), and all variables are given as empirical functions of T and salinity S .
- Note that $\varepsilon_{w\infty}$ in this model differs from the value $\varepsilon_{w\infty} \approx 4.9$ used in the single-Debye model for pure water. Here, $\varepsilon_{w\infty}$ is an empirical function of T and S , ranging from about 3.7 to 4.0 for fresh water ($S = 0$, $0 \leq T \leq 30^\circ\text{C}$) and about 2.8 to 3.1 for typical seawater ($S = 35$ psu). Each value of $\varepsilon_{w\infty}$ belongs to its own model, so the two should not be mixed.
- See Ulaby and Long (2014), Chapter 4, for details.
- The take-away from this model is the simple summation of the effects of multiple mechanisms and the explicit treatment of conductivity in the loss factor (see slide 5 of the lecture slides).
 - Such models are ubiquitous in RS, so many of you will see these things again in your research.

5.3 Freshwater ice

- Let's recall that we can classify the mechanisms that govern permittivity mainly as rotating or stretching.
 - In our capacitor example from before, we illustrated rotation of dipoles, and we just discussed practical, empirical models that consider the characteristic time it takes for dipoles to rotate (relaxation time).
 - It's easy to imagine how dipoles in gas and liquid states can rotate.
 - But solids are distinguished from liquids and gases by having an orderly structure.
 - Water ice has a simple crystalline structure that gives rise to fascinating mechanical properties.
 - * Due to hydrogen bonding, ice (under terrestrial conditions) forms hexagonal crystals.
 - * The bonds that form hexagonal crystals strongly hinder rotation of H_2O molecules in response to an applied electric field, so the molecules can reorient only very slowly.
- Because dipole rotation in ice is so slow, it contributes little at the frequencies used in most RS applications, and models of permittivity in ice often neglect it. For microwave RS applications (the most common for studying ice properties), this effectively means that

$$\varepsilon' \approx 3.2 \quad (\text{virtually independent of } T \text{ and } \omega), \quad \varepsilon'' \approx \frac{\varepsilon_{i0} - \varepsilon_{i\infty}}{\omega\tau_i}, \quad (18)$$

where the variables follow the same basic definitions as for pure water. The approximation for ε'' holds because $\omega\tau_i \gg 1$ at microwave frequencies; the slow rotation of H₂O molecules in ice puts its relaxation frequency in the kilohertz range, so microwave RS sees only the high-frequency tail of this mechanism. This Debye tail dominates ε'' only below about 1–3 GHz. At higher frequencies, an infrared absorption term that increases in proportion to frequency takes over, so that ε'' of ice has a minimum near 1 GHz and rises with frequency above it (Ulaby and Long, 2014, Eq. 4.24 and Fig. 4-3).

- Impurities can be added to this model. See slide 6 in this lecture for more details.

6 Summary

- We just covered a *ton* of material.
- For some of you, this is review with a few new pieces thrown in.
- For some of you, this is almost all new. If you felt lost, ask questions.
- Overview of key take-aways:
 - In RS, we are most interested in the permittivity and conductivity of materials.
 - Conductivity is a measure of a material’s ability to allow electric current (flow of charges: electrons or ions).
 - Permittivity is, for all intents and purposes, a measure of capacitance, i.e., the polarizability of the atoms and molecules that make up the substance.
 - There are multiple polarization mechanisms, most of which involve rotating or stretching and are dominant at different EM frequencies:
 - * *stretching*
 - highest $\omega \rightarrow$ electronic (or atomic) polarization
 - high $\omega \rightarrow$ ionic polarization
 - * *rotation*
 - low $\omega \rightarrow$ dipole rotation
 - * *charge accumulation*
 - lowest $\omega \rightarrow$ interfacial polarization (charge build-up at boundaries)

Variables

Units are SI. Values are given for universal constants; “typical” values are representative, not definitive.

Symbol	Description	Units	Value
<i>Universal constants</i>			
ε_0	permittivity of free space	F/m	8.854×10^{-12}
μ_0	permeability of free space	H/m	$4\pi \times 10^{-7} \approx 1.257 \times 10^{-6}$
c	speed of light in a vacuum, $1/\sqrt{\mu_0\varepsilon_0}$	m/s	2.998×10^8
e	elementary (electron) charge (magnitude)	C	1.602×10^{-19}
k_B	Boltzmann constant	J/K	1.381×10^{-23}

Symbol	Description	Units	Value
j	imaginary unit, $j^2 = -1$ (time dependence $e^{j\omega t}$)	–	–
<i>Constitutive parameters and dielectric properties</i>			
ϵ	complex dielectric constant (relative), $\epsilon' - j\epsilon''$	dimensionless	–
ϵ'	relative permittivity (real part of ϵ)	dimensionless	≥ 1 (almost always); ice ≈ 3.2
ϵ''	dielectric loss factor (negative imaginary part of ϵ)	dimensionless	$\sigma/(\omega\epsilon_0)$ from conduction
$\epsilon_{\text{material}}$	absolute permittivity of a material, $\epsilon'\epsilon_0$	F/m	–
μ	magnetic permeability	H/m	$\approx \mu_0$ in RS
q_v	volume charge density	C/m ³	0 in RS
σ	electrical conductivity	S/m	seawater: ≈ 4.8 (20°C, 35 psu)
ω	angular frequency of the EM wave	rad/s	–
f	frequency, $\omega/2\pi$	Hz	–
<i>Parallel-plate capacitor</i>			
A	plate area	m ²	–
h	plate separation	m	–
Q	charge on each plate	C	–
E	electric field magnitude	V/m	–
$\tilde{E}$	phasor (complex amplitude) of an oscillating field, $E(t) = \text{Re}[\tilde{E}e^{j\omega t}]$	V/m	–
E_+, E_-	fields produced by the positive and negative plates	V/m	$Q/(2A\epsilon_0)$ each
$E_{\text{H}_2\text{O}}$	field of an aligned water dipole	V/m	–
V	potential difference between the plates	V	–
C	capacitance	F	–
<i>Damped harmonic oscillator model</i>			
ξ	particle displacement	m	–
$\tilde{\xi}$	complex amplitude of ξ , $\xi(t) = \text{Re}[\tilde{\xi}e^{j\omega t}]$	m	–
m	mass of the bound particle	kg	–
q	charge of the bound particle	C	–
K_{spring}	effective spring constant of the bond	N/m	$m\omega_0^2$
ω_0	natural (resonant) angular frequency	rad/s	–
ν	damping coefficient	s ⁻¹	–
F_{bond}	restoring (bond) force	N	–
F_{damping}	damping force	N	–
F_{driving}	driving force from the oscillating field	N	–
$p, \tilde{p}$	dipole moment, $q\xi$, and its phasor, $q\tilde{\xi}$	C·m	–
$P, \tilde{P}$	polarization (dipole moment per unit volume), Np , and its phasor	C/m ²	–
N	number of dipoles per unit volume	m ⁻³	–
ω_p	plasma frequency, $\omega_p^2 = Nq^2/(\epsilon_0 m)$	rad/s	–

Symbol	Description	Units	Value
τ_{rel}	characteristic relaxation time of a polarization mechanism	s	$1/\nu$ (free charges); $\approx \nu/\omega_0^2$ (bound, overdamped)
<i>Temperature dependence</i>			
T	temperature	K ($^{\circ}\text{C}$ in empirical fits)	–
K	thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$	–
L	Lorenz number	$\text{W } \Omega \text{ K}^{-2}$	Drude: 1.11×10^{-8} ; Sommerfeld: 2.44×10^{-8}
<i>Water and ice</i>			
ε_{w0}	static (low-frequency) dielectric constant of water	dimensionless	88.0 (0°C), 80.1 (20°C)
$\varepsilon_{w\infty}$	high-frequency dielectric constant of water	dimensionless	4.9 (single-Debye); 2.8–4.0 (double-Debye)
ε_{w1}	intermediate-frequency dielectric constant (double-Debye)	dimensionless	–
τ_w	relaxation time of pure water	s	1.8×10^{-11} (0°C), 9.3×10^{-12} (20°C)
τ_{w1}, τ_{w2}	relaxation times of the two double-Debye terms	s	–
S	salinity	psu (‰)	seawater ≈ 35
$\varepsilon_{i0}, \varepsilon_{i\infty}$	static and high-frequency dielectric constants of ice	dimensionless	–
τ_i	relaxation time of ice	s	relaxation frequency in kHz range